# ANDREW (A.J.) SAGER

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**DATA ANALYTICS**

## PROFESSIONAL SUMMARY

Data analyst with an M.S. in Business Analytics and hands-on experience building KPI frameworks, econometric analyses, ML models, dashboards, and client-ready reports across consulting, energy, media, public-sector, and finance projects. Skilled in Python, SQL, R, Tableau, and Excel. Translates complex datasets into decisions executives can act on. Former Division I athlete.

## WORK EXPERIENCE

**Business Analyst | EcoMetricx** Jan 2026 – Present

- Built a machine learning retention model (LightGBM, Platt-scaled probabilities) to evaluate ~500 streaming catalog titles by their contribution to subscriber churn, then built a budget-constrained optimization that flagged ~17% of the catalog for removal against a $10M/month content budget.
- Modeled release-cadence diminishing returns via simulation and event-study/two-way fixed-effects analyses; estimated willingness-to-pay and price elasticity across three subscriber segments, with predicted cumulative churn landing within 0.1 pp of realized churn.
- Developed KPI scoring and program evaluation analyses for CCA program research, applying PCA and structured rubric design to synthesize quantitative findings into a 185-page strategic report.
- Contributed to 5+ winning RFPs through market research, data analysis, proposal writing, methodology design, and client-facing presentations; finalized multiple discovery-phase reports for public-sector and utility clients.

**Student Capstone Data Analyst | Schneider Electric** Jan 2025 – June 2025

- Built an end-to-end ML forecasting model across four California housing markets, managing data procurement, feature engineering, and model selection; served as project manager and client liaison.

**Financial Planning & Analysis Intern | Sidoxia, LLC** May 2023 – Nov 2023

- Built Excel forecasting models supporting investment decisions for portfolios over $200M; automated reporting via Morningstar and Yahoo Finance integrations, cutting report prep time by 30%.
- Designed 2 predictive investment models; one was adopted into the 2024 portfolio strategy, improving risk-adjusted return analysis.

## TEACHING

**Guest Lecturer** | *ECON 123A: Econometrics, UC Irvine* Nov 2026

- Taught difference-in-differences and applied econometric modeling to undergraduates, along with how AI and LLMs are changing analytical work.

## EDUCATION

**M.S. Business Analytics | UC Irvine, Paul Merage School of Business** June 2025
GPA 3.88 | Beta Gamma Sigma Honor Society (Top 20% of class)

**B.A. Quantitative Economics | UC Irvine** March 2024
GPA 3.70 (3.94 in major)

## SKILLS

**Programming & Analytics:** Python (Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, GeoPandas), R (tidyverse, ggplot2), SQL

**ML & Modeling:** Churn/Survival Modeling, Mathematical Optimization, Classification, Regression, Time Series, PCA, Lasso, Markov Modeling, Simulation, Spatial Modeling

**Econometrics & Analysis:** Event Studies, Difference-in-Differences, Fixed-Effects Modeling, Price-Elasticity/WTP Estimation, Market Sizing, Competitive Analysis, Forecasting, Program Evaluation

**Data & Visualization:** Tableau, Power BI, Excel, MongoDB, ETL Pipelines, Dashboard Development, Leaflet, Git/GitHub

## SELECT PROJECTS

- Climate-Resilient Conservation Prioritization: GIS/ML framework across 4,498 CA planning units combining habitat modeling, climate projections, wildfire history, corridors, optimization, dashboarding, and a Zenodo DOI release (10.5281/zenodo.21880254).
- NCAA Tournament Forecasting Model: XGBoost model with 76% test accuracy vs. a 60% industry benchmark; engineered independent features and validated high-confidence spread calls against Vegas lines.
- San Diego Coffee Shop Analytics Dashboard: public dashboard and scoring system across coffee quality, studyability, parking, price, wait time, service, and proximity.